HR ANALYTICS — PREDICT EMPLOYEE ATTRITION

Internship Project Report | Elevate Labs

1. INTRODUCTION

Employee attrition — the voluntary departure of employees from an organization — is one of the costliest HR challenges in modern business. Replacing a single employee can cost 50–200% of their annual salary, accounting for recruitment, onboarding, and lost productivity.

This project uses data analytics and machine learning on the IBM HR Analytics dataset (1,470 employees, 35 features) to identify the root causes of attrition and build a predictive model that flags at-risk employees before they resign — enabling proactive HR interventions.

2. ABSTRACT

An end-to-end data science pipeline was built covering data cleaning, Exploratory Data Analysis (EDA), ML model training, SHAP explainability analysis, and Power BI dashboard development.

Key Findings:

- **OverTime:** Employees doing OverTime are 3× more likely to leave the company
- **Income:** Lower MonthlyIncome is the second strongest attrition predictor
- **Age & Tenure:** Age group 25–34 and employees under 2 years show highest departure rates
- **Job Role:** Sales Representatives have ~40% attrition — highest of all job roles
- **Model:** Logistic Regression model achieved ~76–82% accuracy with AUC > 0.80

3. TOOLS USED

Tool / Library	Purpose
Python 3 — Google Colab	Cloud-based coding environment — no installation needed
Pandas	Data loading, cleaning, and manipulation (like Excel in Python)
Seaborn / Matplotlib	EDA charts — countplots, boxplots, histograms, heatmaps
Scikit-learn (Sklearn)	Logistic Regression, Decision Tree, accuracy metrics, confusion matrix
SHAP	Model explainability — identifies which features drive attrition
Power BI Desktop	Interactive HR attrition dashboard with slicers and KPI cards
IBM HR Dataset (Kaggle)	Source data — 1,470 employee records, 35 features, zero missing values

4. STEPS INVOLVED IN BUILDING THE PROJECT

#	Step	Details
1	**Data Collection**	Downloaded IBM HR Analytics dataset from Kaggle (free). Contains 1,470 employee records with 35 features including demographics, compensation, satisfaction scores, and attrition status.
2	**Data Cleaning & EDA**	Verified zero missing values and zero duplicates. Analyzed attrition patterns across Department, Job Role, Age, Income, OverTime, and Job Satisfaction using 5 EDA charts.

3	**Preprocessing**	Removed 4 non-informative columns. Applied LabelEncoder to convert 9 text columns to numbers. Applied StandardScaler to normalize features. Split 80/20 into train/test sets.
4	**Model Training**	Trained Logistic Regression with class_weight='balanced' (handles 16%/84% imbalance) and Decision Tree with max_depth=5. Compared models using Accuracy, AUC-ROC, and Classification Report.
5	**SHAP Analysis**	Applied SHAP LinearExplainer to identify which features most influenced each prediction. Generated dot summary plot and top-10 bar chart showing OverTime, Income, and Age as top drivers.
6	**Power BI Dashboard**	Built 6-visual dashboard: KPI cards (total, left, rate%), pie chart, department/role bar charts, age group chart. Added slicers for Department, Gender, and OverTime filtering.
7	**PDF Report**	Compiled findings, charts, and recommendations into this structured 2-page internship report following Elevate Labs submission guidelines.

5. ATTRITION PREVENTION SUGGESTIONS (from SHAP Analysis)

Factor	Finding	Action Recommended
OverTime	Employees with OverTime = Yes are 3x more likely to leave the company	Cap mandatory overtime at 5 hrs/week. Offer compensatory off or extra pay for overtime hours
Monthly Income	Employees earning below department median have 28% higher attrition probability	Conduct annual salary benchmarking against industry standards. Review pay bands for Sales roles
Job Satisfaction	Level 1 satisfaction has 22.8% attrition vs 11.3% at Level 4	Quarterly pulse surveys. Manager one-on-one check-ins every 4–6 weeks for low-satisfaction employees
Years at Company	Attrition peaks in first 2 years — critical onboarding risk window	Build a structured 90-day and 1-year onboarding and mentorship program for all new joiners
Age Group (25–34)	Younger employees seek faster career growth and leave if not progressing	Create visible career roadmaps and promotion criteria for early-career staff within their first year

6. CONCLUSION

This project successfully demonstrated how data analytics and machine learning can transform HR decision-making from reactive to proactive. By applying Logistic Regression and SHAP analysis to the IBM HR dataset, it was possible to:

- Identify OverTime, MonthlyIncome, Age, and JobSatisfaction as the top four attrition drivers
- Build a classification model achieving ~76–82% accuracy with AUC > 0.80
- Generate six actionable business recommendations directly tied to SHAP feature evidence
- Visualize attrition trends interactively in a Power BI dashboard for non-technical HR stakeholders

The skills applied — end-to-end data pipeline, class imbalance handling, model explainability with SHAP, and stakeholder-facing dashboards — are directly aligned with the core competencies expected of a Data Analyst. This project validates the ability to translate raw HR data into measurable business impact, meeting the Elevate Labs internship objective of building interview-ready portfolio work.